

Temperature sensor

DETECT

INPUT

ANALOGUE

What does it do?

A TMP36 turns temperature into a changing voltage that a program can measure.

Its output rises by 10 millivolts for every degree Celsius, with 0.5 V representing 0 °C.

The Pico reads that voltage directly; the Raspberry Pi uses an MCP3008 analogue-to-digital converter.

Uses



Refrigerator cooling

DETECT
Measure
fridge
temperature

DECIDE
Is it too
warm?

DO
Increase the
cooling

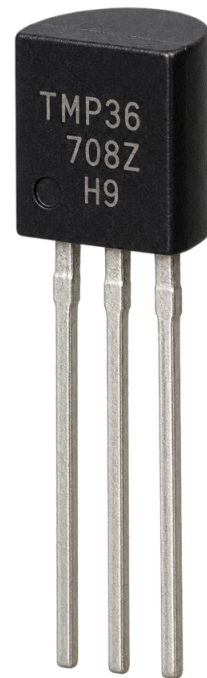


Soup drama meter

DETECT
Measure bowl
surface

DECIDE
Which drama
level fits?

DO
Move the
drama pointer



**Flat
face**

Use it to
identify
the pins

+Vs
Left pin

Vout
Centre pin

GND
Right pin

Code

● Raspberry Pi

```
from gpiozero import MCP3008

sensor = MCP3008(channel=0)

temperature = (sensor.voltage - 0.5) * 100
```

● Raspberry Pi Pico

```
from picozero import TemperatureSensor

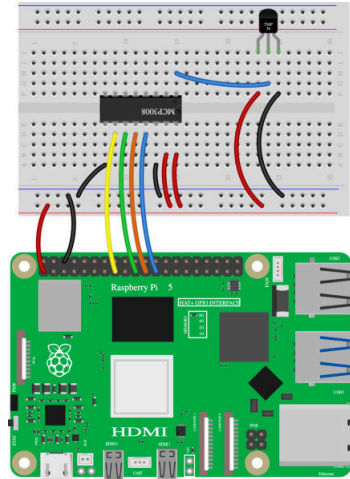
convert = lambda voltage: (voltage - 0.5) * 100
sensor = TemperatureSensor(26, conversion=convert)

temperature = sensor.temp
```

Read the TMP36 with its flat face towards you; the pins are +Vs, Vout, GND from left to right.

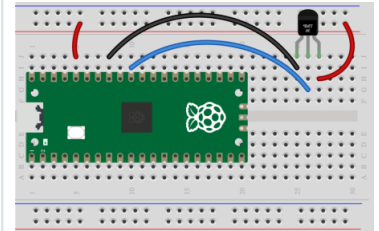
Wiring

RASPBERRY PI 5



TMP36 **Vout** → MCP3008 **CHO**; ADC SPI uses GPIO **22, 10, 9, 11**

PICO



TMP36 **+Vs** → 3V3 OUT, **Vout** → GP26, **GND** → GND